

Revised Simplex method (Standard form-II)

Big-M Method

$$\left\{ \begin{array}{l} \text{Min } Z = 4x_1 + 3x_2 \\ \text{Subject to } x_1 + 2x_2 \geq 8 \\ 3x_1 + 2x_2 \geq 12 \\ x_1, x_2 \geq 0 \end{array} \right\}$$

$$\boxed{\text{Min } Z = - \text{Max } (-Z)}$$

$$\Rightarrow \text{Max } (-Z) = - \text{Min } Z$$

$$\Rightarrow \text{Max } Z' = - \text{Min } Z$$

$$\text{Let } Z' = -Z$$

$$\text{Max } Z' = -4x_1 - 3x_2 \quad \text{--- (1)}$$

Subject to

$$\left\{ \begin{array}{l} x_1 + 2x_2 - x_3 + x_5 = 8 \\ 3x_1 + 2x_2 - x_4 + x_6 = 12 \\ x_1, x_2, \dots, x_6 \geq 0 \end{array} \right\}$$

$x_3, x_4 \rightarrow$ Surplus variables
and $x_5, x_6 \rightarrow$ artificial variables

$$\text{Max } Z' = -4x_1 - 3x_2 + 0 \cdot x_3 + 0 \cdot x_4 - \underline{M} \cdot x_5 - M \cdot x_6 \rightarrow$$

$$\Rightarrow \left\{ \begin{array}{l} Z' + 4x_1 + 3x_2 + 0 \cdot x_3 + 0 \cdot x_4 + M \cdot x_5 + M \cdot x_6 = 0 \\ x_1 + 2x_2 - x_3 + 0 \cdot x_4 + x_5 + 0 \cdot x_6 = 8 \quad \text{--- (1)} \\ 3x_1 + 2x_2 + 0 \cdot x_3 - x_4 + 0 \cdot x_5 + x_6 = 12 \quad \text{--- (2)} \end{array} \right. \quad \text{--- (3)}$$

$$\begin{array}{c}
 z' \quad x_1 \quad x_2 \quad x_3 \quad x_4 \quad x_5 \quad x_6 \\
 \begin{bmatrix} 1 & 4 & 3 & 0 & 0 & M & M \\ 0 & 1 & 2 & -1 & 0 & 1 & 0 \\ 0 & 3 & 2 & 0 & -1 & 0 & 1 \end{bmatrix} \begin{bmatrix} z' \\ x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \end{bmatrix} = \begin{bmatrix} 0 \\ 8 \\ 12 \end{bmatrix}
 \end{array}$$

$\begin{matrix} \uparrow \\ a_0^* \end{matrix}$
 $\begin{matrix} \uparrow \\ a_1^* \end{matrix}$
 $\begin{matrix} \uparrow \\ a_2^* \end{matrix}$
 $\begin{matrix} \uparrow \\ a_3^* \end{matrix}$
 $\begin{matrix} \uparrow \\ a_4^* \end{matrix}$
 $\begin{matrix} \uparrow \\ a_5^* \end{matrix}$
 $\begin{matrix} \uparrow \\ a_6^* \end{matrix}$

$b^* = \begin{bmatrix} 0 \\ b \end{bmatrix}$
 where $b = \begin{bmatrix} 8 \\ 12 \end{bmatrix}$

$\rightarrow (a_0^*, a_5^*, a_6^*) \rightarrow \text{Basis.}$

$$B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad B^* = \begin{bmatrix} 1 & -C_B \\ 0 & B \end{bmatrix}$$

If $C_B = 0 \Rightarrow B^* = \text{Identity matrix.}$

$$C_B = (-M, -M)$$

$$B^* = \begin{bmatrix} 1 & M & M \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \Rightarrow B^{*-1} = \begin{bmatrix} 1 & C_B B^{-1} \\ 0 & B^{-1} \end{bmatrix}$$

$$= \begin{bmatrix} 1 & -M & -M \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Iteration - I

Revised Simplex Table

Basic variable	B^{*-1}			x_B	x_k	x_B/x_k
	β_0	β_1	β_2			
z'	1	-M	-M	-20M	-4M+3	$\frac{20M}{4M-3}$
x_5	0	1	0	8	2	$8/2=4$
x_6	0	0	1	12	2	$12/2=6$

$$x_B^* = B^{*-1} b^* = \begin{bmatrix} 1 & -M & -M \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 8 \\ 12 \end{bmatrix}$$

$$= \begin{bmatrix} -20M \\ 8 \\ 12 \end{bmatrix}$$

Additional Table

a_1^*	a_2^*	a_3^*	a_4^*
4	3	0	0
1	2	-1	0
3	2	0	-1

Optimality condition

$$\Delta = [\text{First row of } B^{*-1}] \times \text{Additional Table}$$

$$= [1, -M, -M] \times \begin{bmatrix} 4 & 3 & 0 & 0 \\ 1 & 2 & -1 & 0 \\ 3 & 2 & 0 & -1 \end{bmatrix}$$

$$= (-4M+4, -4M+3, M, M)$$

check $\Delta_j \geq 0 \quad \forall j$

Δ_1, Δ_2 are -ve value.

$$\underline{\underline{\text{Min } (\Delta_j)}} = \text{Min}(-4M+4, -4M+3, M, M) \\ = -4M+3 = \Delta_2$$

$K=2 \Rightarrow \underline{\underline{a_2^*}}$ is entering to next basis

$$\boxed{y_2 = B^{-1} a_2}$$

$$\underline{\underline{y_2^*}} = B^{*-1} a_2^* = \begin{bmatrix} 1 & -M & -M \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \\ 2 \end{bmatrix} \\ = \begin{bmatrix} -4M+3 \\ 2 \\ 2 \end{bmatrix}$$

$$\text{Min} \left(\frac{x_B}{x_K} \right) = \text{Min} \left(\frac{20M}{4M-3}, 4, 6 \right) \\ = 4$$

$\Rightarrow \underline{\underline{x_5}}$ is leaving variable.

$\Rightarrow 2$ is the key element.

Improved co-efficient

B_1	B_2	X_B	X_K
$-M$	$-M$	$-20M$	$-4M+3$
1	0	8	2 ←
0	1	12	2

B_0

	B_1	B_2	X_B	X_K
1	$M - \frac{3}{2}$	$-M$	$-4M - 12$	0
0	$\frac{1}{2}$	0	4	1
0	-1	1	4	0

$$R_1 \leftarrow R_1 - (-4M+3)R_2$$

$$R_2 \leftarrow \frac{1}{2} R_2$$

$$R_3 \leftarrow R_3 - 2R_2$$

Iteration - II

B.V	B^{*-1}			X_B	X_K	X_B / X_K
	B_0	B_1	B_2			
Z'	1	$M - \frac{3}{2}$	$-M$	$-4M - 12$	$-2M + \frac{5}{2}$	—
x_2	0	$\frac{1}{2}$	0	4	$\frac{1}{2}$	8
x_6	0	-1	1	4	2	$\frac{4}{2}$ ←

Additional Table

a_1^*	a_3^*	a_4^*
4	0	0
1	-1	0
3	0	-1

Optimality condition:

$$\begin{aligned}\Delta &= (\text{First row of } B^{*-1}) \times \text{Additional Table} \\ &= (1, M - \frac{3}{2}, -M) \times \begin{pmatrix} 4 & 0 & 0 \\ 1 & -1 & 0 \\ 3 & 0 & -1 \end{pmatrix} \\ &= \left(\boxed{-2M + \frac{5}{2}}, \underline{-M + \frac{3}{2}}, M \right)\end{aligned}$$

check $\Delta_j \geq 0 \quad \forall j$ or not??

Entering variable

$$\min(\Delta_j)$$

$$= \min\left(-2M + \frac{5}{2}, -M + \frac{3}{2}, M\right)$$

$\uparrow$

$\Rightarrow a_1^*$ will enter to the new basis

Leaving variable:

$$\begin{aligned}y_1^* &= B^{-1} a_1 = \begin{bmatrix} \frac{1}{2} & 0 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 3 \end{bmatrix} \\ &= \begin{bmatrix} \frac{1}{2} \\ 2 \end{bmatrix}\end{aligned}$$

$$y_1^* = \begin{bmatrix} -2M + 5/2 \\ \frac{1}{2} \\ 2 \end{bmatrix}$$

$\Rightarrow x_6$ is the leaving variable

Improved coefficient

B_1	B_2	X_B	X_K
$M - 3/2$	$-M$	$-4M - 12$	$-2M + 5/2$
$1/2$	0	4	$1/2$
-1	1	4	2

B_1	B_2	X_B	X_K
$-1/4$	$-5/4$	-17	0
$3/4$	$-1/4$	3	0
$-1/2$	$1/2$	2	1

$$R_1 \leftarrow R_1 - (-2M + 5/2) R_3$$

$$R_3 \leftarrow \frac{1}{2} R_3$$

$$R_2 \leftarrow R_2 - \frac{1}{2} R_3$$

Iteration - III

Basic variables	B^{*-1}			x_B	x_R
	B_0	B_1	B_2		
Z'	1	$-\frac{1}{4}$	$-\frac{9}{4}$	-17	
x_2	0	$\frac{3}{4}$	$-\frac{1}{4}$	3	
x_1	0	$-\frac{1}{2}$	$\frac{1}{2}$	2	

$$x_B^* > \underline{\underline{B^{*-1} b^*}} \quad \text{check??}$$

optimality

$$\Delta = (\text{first row of } B^{*-1}) \times$$

(Additional Table)

$$= (1, -\frac{1}{4}, -\frac{9}{4}) \times \begin{bmatrix} 0 & 0 \\ -1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$= \left(\frac{1}{4}, \frac{5}{4} \right)$$

$$\text{As } \Delta_j \geq 0 \quad \forall j$$

$\Rightarrow$ optimality arises.

$$\begin{array}{c|c} \text{Max } Z' = -17 & x_1 = 2 \\ & x_2 = 3 \end{array}$$

$\downarrow$

$$\text{Min } Z = -\text{Max}(-Z)$$

$$= -\text{Max } Z'$$

$$= -(-17)$$

$$\boxed{= 17}$$

$$\text{Min } Z = 17 \quad \text{for}$$

$$x_1 = 2, x_2 = 3$$

checking $x_1 = 2, x_2 = 3$.

$$\begin{aligned} \text{Min } Z &= 4x_1 + 3x_2 \\ &= 4 \times 2 + 3 \times 3 \end{aligned}$$

$$= 8 + 9$$

$$= 17$$